Distribution Maps of Early Neolithic in Central Europe and in Rhineland

J. Hilpert, J. Meesenburg, K.P. Wendt, A. Zimmermann

Abstract

The distribution of early Neolithic Bandkeramik sites is represented in two scales (central Europe and Rhineland) by detailed point distribution patterns and generalized polygons of high density areas. Location of sites is derived by digitizing published maps.

Data: for central Europe from PREUSS (ed.) 1998 Map 1;
for Rhineland from RICHTER / CLAßEN 1997

The **method** to delimit high density areas in a standardized way follows a protocol with six steps (see below point 4). This combination of methods is published e. g. by Hilpert et al. 2008 and Zimmermann et al. 2009. More details in Zimmermann et al. 2004.

Scientific questions: These maps were used to estimate population densities for different periods of time (most recent reference Wendt et al. 2010). In case of the early Neolithic the optimal isoline is chosen by the 4km radius of the LEC for the scale level Rhineland; for central Europe the optimal isoline is chosen by the 3.5 km radius of the LEC.

Web publication:

1. Shape files of point distribution patterns for central Europe in two projections. Access to high precision coordinates at a scale such as Rhineland allowing localisation of archaeological sites in the terrain is limited due to protection reason.
2. Shape files of polygons of high site densities for Rhineland and central Europe in two projections each.
3. Spreadsheet to document selection of optimal isoline delimiting high density areas for Rhineland and central Europe.

1 Context

Collecting data and creating GIS datasets was conducted within the Rhine-LUCIFS-Project funded by German Research Foundation from 2002 until 2017 (http://ufg.phil-fak.uni-koeln.de/36039.html). LUCIFS is part of the PAGES framework.

LUCIFS = Land Use and Climate Impact on Fluvial Systems during the period of agriculture

PAGES = Past Global Changes

The dataset is assigned with a DOI, and can be cited as follows in scholarly works:

> J. Hilpert, K.P. Wendt, J. Meesenburg, A. Zimmermann 2019: Rhine-LUCIFS-Project: Sites and settlement areas of the early Neolithic of the scale levels Rhineland and Germany. Rhine-LUCIFS-Database, doi: ???/SFB806.20.

2 Metadata

The basic descriptive metadata of the dataset is given in this section.

2.1 Basic metadata

Title	Distribution Maps of Early Neolithic in Central Europe and in Rhineland
Author(s)	J. Hilpert, J. Meesenburg, K.P. Wendt, A. Zimmermann
Year	2019
Topic	Sites, Settlement areas
Keywords	Stone Age, early Neolithic, sites, settlement areas
Publisher	Rhine-LUCIFS-Database
DOI	?

2.2 Spatial metadata

Region 1	Rhineland (Rhl)
Type	Bounding box
Bounding box 1 (SW, NE)	2508225/5466669; 2651733/5667303 (EPSG:31466) *6.113/49.335; 8.106/51.140 (EPSG:4326)*
Projection (EPSG) = unit	EPSG: 31466 = meter *EPSG: 4326 = grad*
Region 2	Central Europe (CE)
Bounding box 2 (SW, NE)	3254015/5268458; 4000605/5929919 *5.527/47.546; 16.166/53.397*
Projection (EPSG) = unit	EPSG: 31467 = meter *EPSG: 4326 = grad*

2.3 Temporal metadata

Type	Interval
Name	Early Neolithic: Linearbandkeramik
Interval begin	7500 cal. BP
Interval end	6950 cal. BP

3 Data sources

The final datasets are a collection of vector data selected from published archaeological maps for the areas of interest. The Rhineland data is based on J. RICHTER / E. Claßen 1997, Neolithikum. Geschichtlicher Atlas Rheinlande Beih. II/2 Karte 1: Alt- und Mittelneolithikum (Köln 1997). The data set was updated with data from the Cultural Heritage administration in Bonn. Access to high precision location of archaeological sites is limited due to protection reasons. Therefore, only settlement areas are chosen for Web publication. The data from the scale level Germany is based on J. Preuss (ed.) 1998. Das Neolithikum in Mitteleuropa: Kulturen, Wirtschaft, Umwelt vom 6. bis 3. Jahrtausend v.u.Z., Übersichten zum Stand der Forschung, Karte 1: Verbreitung neolithischer Kulturen in Mitteleuropa (Weissbach 1998). Of several versions of isolines the one of J. Meesenburg in Schlummer et al. 2014, 38-39 was chosen.

Published dataset	Spatial	Source
Early Neolithic sites	Rhineland	J. RICHTER / E. Claßen 1997, Karte 1: Alt- und Mittelneolithikum.
Early Neolithic sites	Central Europe	J. Preuss (ed.) 1998. Das Neolithikum in Mitteleuropa: Kulturen, Wirtschaft, Umwelt vom 6. bis 3. Jahrtausend v.u.Z., : Verbreitung neolithischer Kulturen in Mitteleuropa (Weissbach 1998) Karte 1
	Germany	Zimmermann et al. 2004, Abb. 15; Zimmermann et al- 2009, Fig. 4

4 Data set creation and provenance

The datasets have been created by digitizing the archaeological sites from scanned maps. Based on these points, maps of site densities have been created by a combination of six steps:

1. Calculate Thiessen polygons based on site location
2. Determine radius of "Largest Empty Circles" (LEC) based on nodes of Thiessen polygons (PREPARATA/SHAMOS 1985) to calculate site densities
3. Interpolate site densities between locations of nodes of Thiessen polygons by kriging (HAAS/VIALLIX 1976)
4. Derive isolines for areas of the same minimum density
5. Calculate increase of area for succeeding isolines starting in areas of highest densities
6. Select isoline as optimal where increase of area is maximal. The result is most sensible to this selection, especially if a local maximum has to be chosen. Therefore, the spreadsheet is included into the documentation which was used to derive this decision.

Various GIS-software is able to apply the described methods (ArcInfo, CartaLinx, MapInfo, Idrisi,).

5 Maps and visualisations

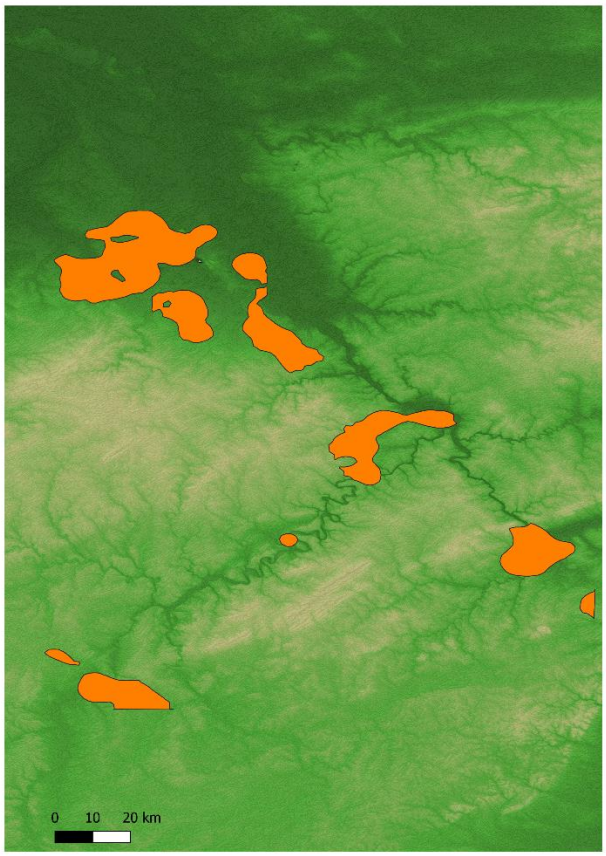


Figure 1: Settlement areas of the early Neolithic in the Rhineland (isoline of the 4km LEC-radius).

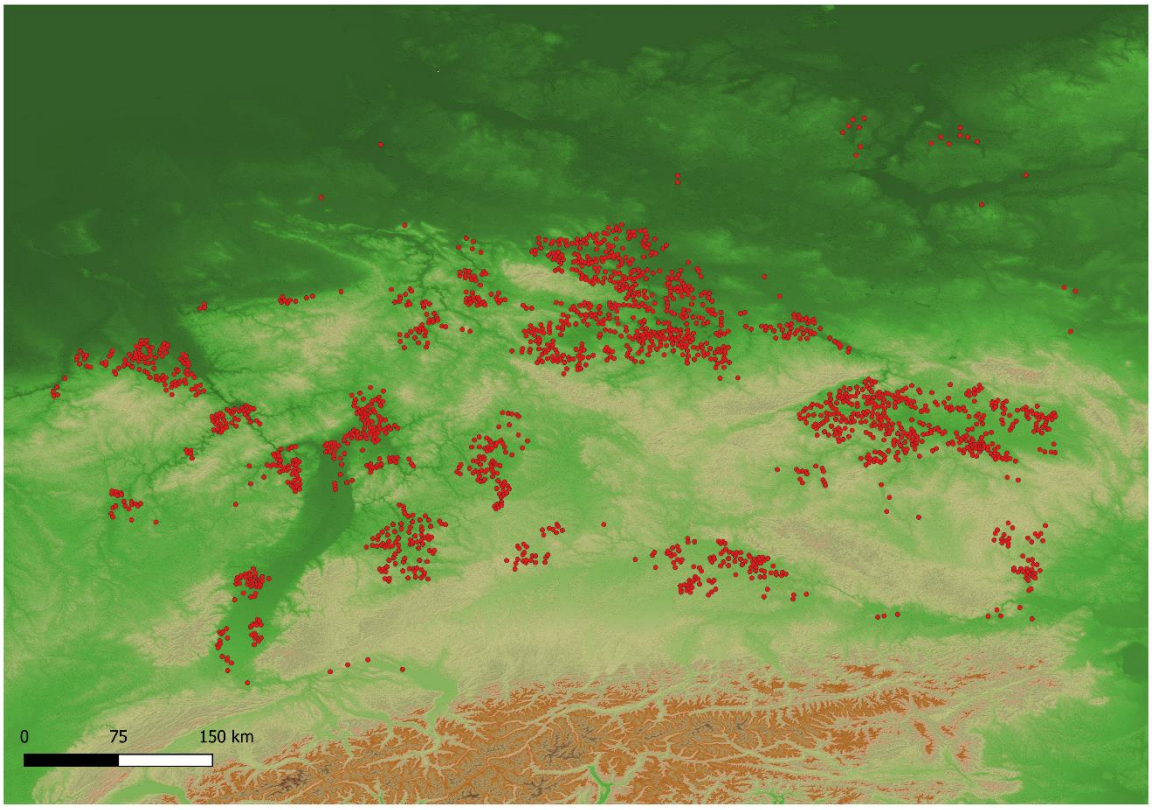


Figure 2: Early Neolithic sites in Central Europe after Preuss (ed.) 1998.

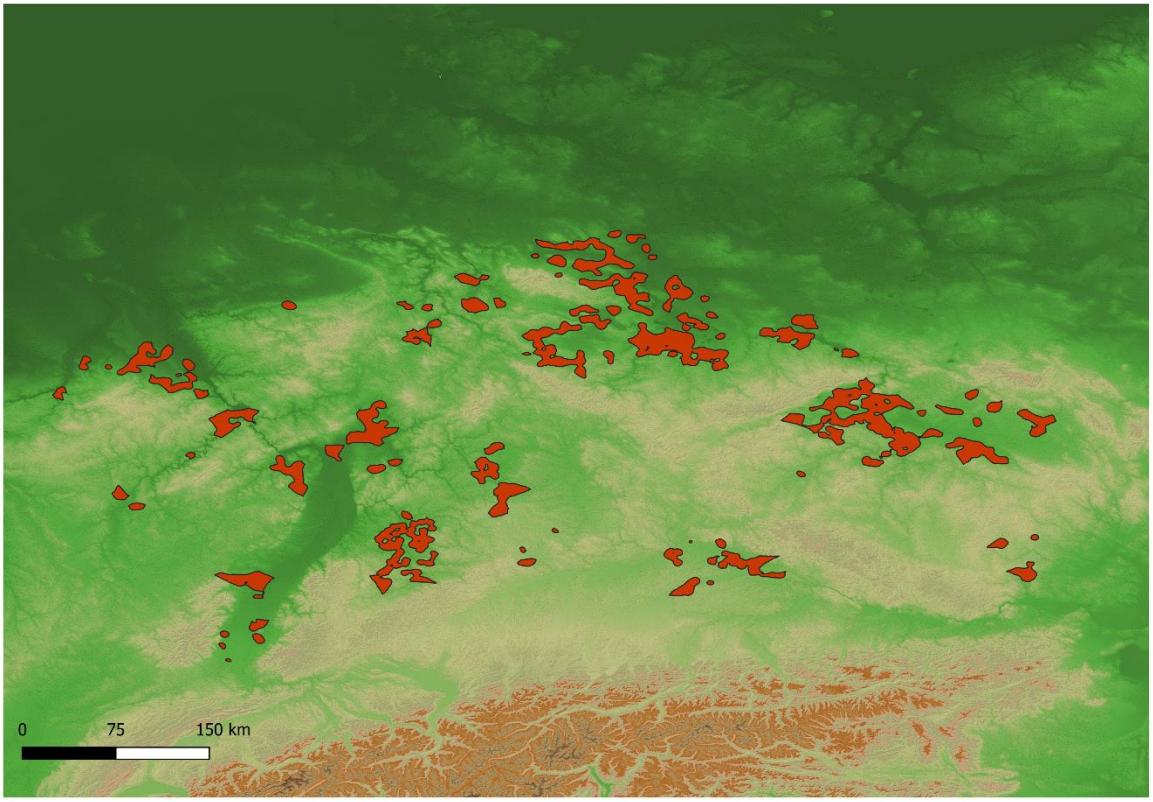


Figure 3: Settlement areas of the early Neolithic in Central Europe (isoline of the 3,5km LEC-radius), derived from sites after Preuss (ed.) 1998 (see Figure 2).

The figures are showing visualisations of the available datasets in relation to the topography of the region created with open source geodata: A.JARVIS / H.I. REUTER / A. NELSON / E. GUEVARA, Hole-filled seamless SRTM data V4, International Centre for Tropical Agriculture (CIAT) 2008, available from http://srtm.csi.cgiar.org.

6 Data resources

6.1 File resources

Polygon vector data (settlement areas)				
Name	Pro-jection	Objects	File-type	Size
13_earlyNeolithic_CE_iso3,5km_GK3.zip	EPSG: 31467	N=91	Zipped Shapefile	177.932 KB
13a_earlyNeolithic_Rhl_Iso4km_GK2.zip	EPSG: 31466	N=10	Zipped Shapefile	7.991 KB
13_earlyNeolithic_CE_iso3,5km_WGS84.ZIP	EPSG: 4326	N=91	Zipped Shapefile	249.527 KB
13a_earlyNeolithic_Rhl_iso4km_WGS84.ZIP	EPSG: 4326	N=10	Zipped Shapefile	11.460 KB
Point vector data				
Name	Pro-jection	Objects	File-type	Size
13_earlyNeolithic_CE_sites_GK3.zip	EPSG 31467	N=2378	Zipped Shapefile	93.022 KB
13_earlyNeolithic_CE_sites_WGS84.ZIP	EPSG: 4326	N=2378	Zipped Shapefile	115.970 KB
Excel charts				
Name			File-type	Size
13_earlyNeolithic_CE.xls	-	-	MS Excel file	33.792 KB
13a_earlyNeolithic_Rhl.xls	-	-	MS Excel file	55.296 KB
Text				
Name			File-type	Size
13_early_Neolithic_CE_Metadata.docx	-	-	PDF	68.348 KB

6.2 Web resources

DOI	**???/SFB806.41**

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